Name: ______________________ Date: ______________________

SLIDE 1

NOTES

Class Agenda

01 **Stress & Strain**
The fundamental language for describing how metals respond to force

02 **Crystal Defects**
Point, line, planar, and volume defects — and why they matter more than you'd expect

03 **Grain Boundaries & Diffusion**
How boundaries strengthen metal, the Hall-Petch relationship, and how atoms move through solid metal

SLIDE 2 *Class Agenda*

NOTES

Metallography Lab — Module 2

EXAMET-5-R-600-HD · 50–500×

Module 2 · Week 2

Mechanical Properties

What to do this week:

Compare pearlite fraction in A vs B. Measure the Vickers indent diagonal (now on B) and calculate HV.

Steps:

1 Observe — complete feature table
2 Sketch — label magnification & specimen
3 Answer — 4 short questions

Specimens to Check Out

· Specimen A — 1018 steel

· Specimen B — 1045 steel (Vickers indent applied)

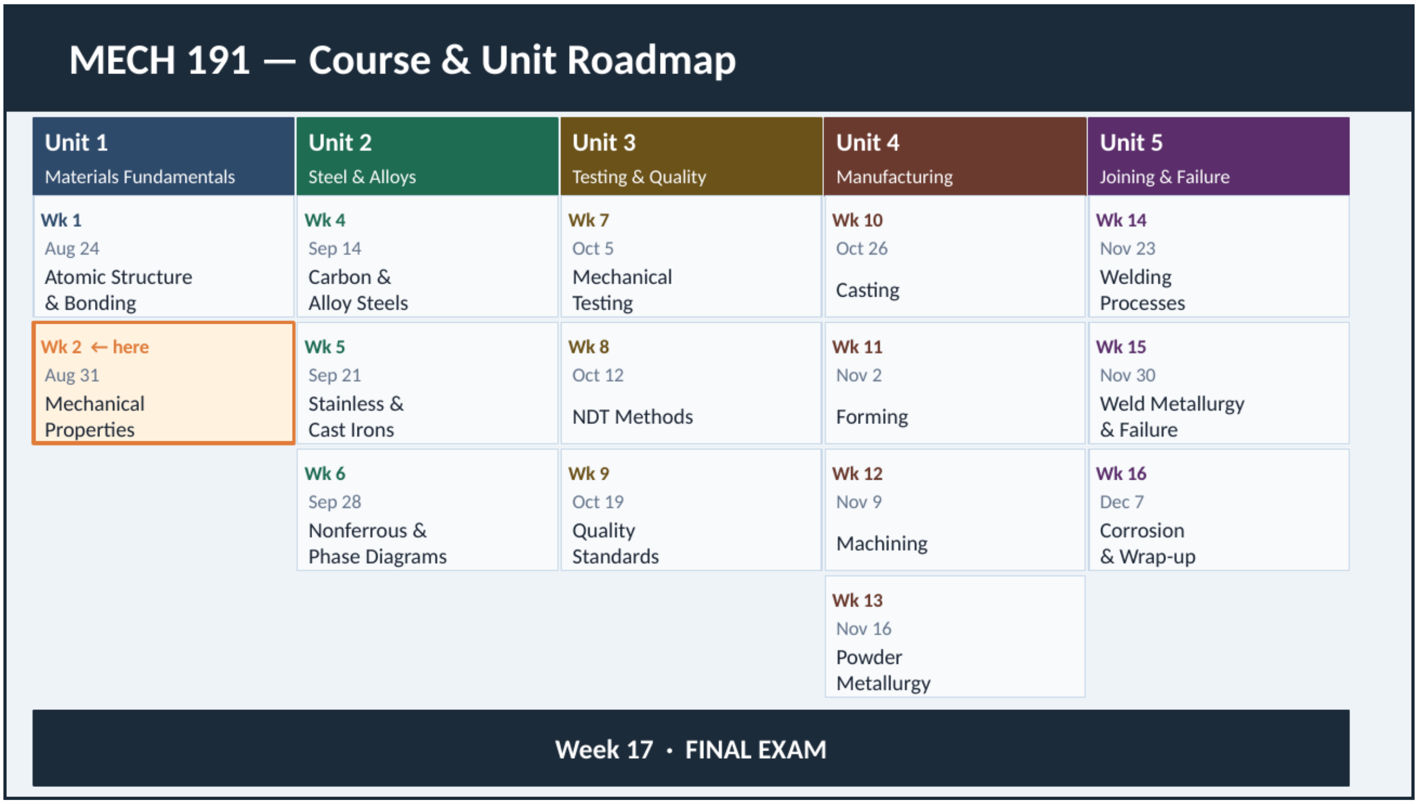
How to check out the kit:

1 See instructor after class today
2 Sign out the specimen tray
3 Use the EXAMET-5 at your convenience
4 Return kit before leaving class
5 Submit completed module with your portfolio

Portfolio Handout — Module 2

SLIDE 3 *Metallography Lab — Module 2 EXAMET-5-R-600-HD · 50–500×*

NOTES



SLIDE 4 *MECH 191 — Course & Unit Roadmap*

NOTES

Week 2 — At a Glance

Topics

- Stress and strain — the language of mechanical behavior
- Elastic vs. plastic deformation and the yield point
- Key properties: strength, hardness, ductility, toughness, brittleness
- Crystal defects — point, line, planar, and volume
- Grain boundaries — formation, energy, and the Hall-Petch effect
- Diffusion — how atoms move through solid metal

Resources

Reference Material

Moniz 5th — Ch. 3
Mechanical Properties of Metals

Kalpakjian 6th — Ch. 2
Fundamentals of Mechanical Behavior of Materials

Visualizers

- Stress-Strain Curve (Interactive)
- Hot vs Cold Working & Work Hardening

SLIDE 5

NOTES

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

1. What is the difference between strength and toughness? Can a material be strong but not tough?
2. At the atomic level, what exactly happens at the yield point when you pull a metal specimen in tension?
3. Cold-working a metal (rolling, drawing) increases its strength but reduces ductility. What microstructural change explains both effects simultaneously?
4. A fine-grained and a coarse-grained steel have identical composition. Which yields at higher stress — and why?
5. A steel shaft fails well below its yield strength after thousands of rotation cycles. What failure mode is this, and what property should have been checked?

SLIDE 6 *Core Questions*

NOTES

Stress and Strain — The Language

When a force is applied to a metal, we need precise language to describe what happens inside it — at the grain level and at the atomic level.

Stress (σ)

$$\sigma = F / A$$

Internal force per unit area — the pressure the atoms inside the material experience when an external load is applied.

Types of stress:

Tensile — pulling apart (positive σ)
Compressive — pushing together (negative σ)
Shear — sliding one plane over another (τ)

Units: Pa, MPa, ksi

Strain (ε)

$$\varepsilon = \Delta L / L_0$$

The amount of deformation relative to original size — how much the material has stretched or compressed under load.

Key points:

Dimensionless — or expressed as %
Ductility is reported as % elongation at fracture
Strain rate matters — fast loading vs. slow loading

Dimensionless — expressed as mm/mm, in/in, or %

SLIDE 7 *Stress and Strain — The Language*

NOTES

Elastic vs. Plastic Deformation

Elastic Zone

The material deforms under load but returns to its original shape when the load is removed — like a spring. Atoms are displaced but not permanently rearranged. The bonds stretch, then pull back.

Example:

Spring, tight bolt that can still be removed, a bent ruler that snaps back

Yield Point

The critical transition stress — above this, deformation becomes permanent. Atoms and dislocations have moved too far to recover. This is the most important value in material selection for structural design.

Example:

The 'click' when a bolt is overtightened — it won't come back to correct preload

Plastic Zone

Permanent deformation — the material will not spring back. Dislocations are moving through the crystal, allowing shape change without fracture. The material flows, distributing stress.

Example:

Bent paperclip, cold-formed bracket, stamped sheet metal door panel

Ultimate Tensile Strength (UTS): the maximum stress the material can handle before fracture — the peak of the stress-strain curve, past the yield point.

SLIDE 8 *Elastic vs. Plastic Deformation*

NOTES

Key Mechanical Properties

These five properties are how engineers describe and compare materials — you will encounter all of them in specifications, data sheets, and test reports.

σ Strength	H Hardness	% Ductility	J Toughness	✗ Brittleness
Maximum stress a material can handle before yielding (yield strength) or fracturing (UTS). Higher = harder to permanently deform or break.	Resistance of the surface to indentation or scratching. Directly related to yield strength — harder materials resist surface deformation.	How much plastic deformation a material can tolerate before fracturing. Reported as % elongation. Essential for forming operations.	Energy absorbed before fracture — the area under the stress-strain curve. High toughness = survives impacts. Requires both strength and ductility.	Fractures with little or no plastic deformation. Brittle materials can be very strong but fail suddenly with no warning. The opposite of toughness.
How tested: Tensile test, compression test	**How tested:** Rockwell, Brinell, Vickers	**How tested:** % elongation, % area reduction	**How tested:** Charpy, Izod impact tests	**How tested:** Observed at the fracture surface

SLIDE 9 *Key Mechanical Properties*

NOTES

Crystal Defects — Point Defects

No real metal is perfect — defects exist at every scale. Point defects are disruptions at a single lattice site. Every cubic centimeter of real metal contains millions.

Vacancy

A missing atom — an empty site where an atom should be. The most common point defect in all metals. Number increases exponentially with temperature.

Why it matters:

Enable atom movement (diffusion). Essential for heat treatment, annealing, and solid-state reactions.

One missing atom out of billions — but millions of them create measurable softening and diffusion pathways.

Interstitial Defect

An extra atom squeezed into a gap between host atoms. If the same element: self-interstitial. If a smaller foreign element: interstitial alloy.

Why it matters:

Carbon in iron is an interstitial defect — even tiny amounts dramatically harden steel by blocking dislocation movement.

This is exactly what we covered in Week 1 — the mechanism behind carbon steels.

Substitutional Defect

A wrong atom sitting in a lattice site, replacing a host atom. Works best when the substitute is within ~15% in size and has similar valence electrons.

Why it matters:

Cr, Ni, Mo in steel are all substitutional — this is how stainless steel and alloy steels are made.

Same mechanism as substitutional solid solutions from Week 1.

SLIDE 10 *Crystal Defects — Point Defects*

NOTES

Crystal Defects — Line Defects (Dislocations)

Core question:
If dislocations make metals easier to deform, why don't we want to eliminate them? A metal with zero dislocations would need enormous stress to deform at all, then fracture suddenly with no warning — like glass.

Edge Dislocation

An extra half-plane of atoms inserted into the lattice — like forcing an extra card into a deck halfway through. The distortion runs along the edge of this extra plane.

Burgers vector (b): perpendicular to the dislocation line

Like a wrinkle in a rug — slide the wrinkle across instead of lifting the whole rug at once

Screw Dislocation

A helical distortion of the lattice — atomic planes spiral around the dislocation line like a corkscrew. Can cross-slip between slip planes, giving more paths to deform.

Burgers vector (b): parallel to the dislocation line

Like tearing paper by sliding the two halves sideways rather than pulling them apart

Mixed Dislocation

In practice, most dislocations are a mixture of edge and screw character — the Burgers vector is at an angle. Pure edge or pure screw are rarely found in real metals.

Burgers vector: at an angle — has both edge and screw components

Real metal dislocations wind and curve through the crystal in complex three-dimensional paths

SLIDE 11 *Crystal Defects — Line Defects (Dislocations)*

NOTES

Crystal Defects — Planar Defects

Two-dimensional surfaces where crystal structure or orientation changes — grain boundaries are the most common and most important.

Grain Boundaries

The most common planar defect. When independently nucleating crystals grow into each other during solidification, the orientation mismatch creates a disordered boundary region.

- High misorientation (>15°): high disorder, high energy, more reactive to corrosion and diffusion
- Low misorientation (<15°): tilt or twist boundary — more structured, lower energy

Twin Boundaries

A boundary where the crystal on each side is a perfect mirror image of the other — extremely low energy and very stable. Two types with different origins:

- Annealing twins: form during recrystallization of FCC metals (Cu, Al, austenitic stainless steel)
- Deformation twins: form under shock loading or low-temperature plastic deformation — common in HCP and BCC

Stacking Faults

A disruption in the normal atomic stacking sequence — FCC follows ABCABC... and HCP follows ABAB...; a fault breaks this pattern. Low Stacking Fault Energy (SFE) → wide stable faults → dislocations can't cross-slip → the metal work-hardens quickly. High SFE → narrow faults → easy cross-slip → softer cold-working behavior.

SLIDE 12 *Crystal Defects — Planar Defects*

NOTES

Crystal Defects — Volume Defects

Three-dimensional imperfections large enough to detect with inspection equipment. These are the defects that cause failures in service.

Voids & Porosity

Empty spaces trapped during solidification.

- Gas porosity: dissolved gases escape as metal solidifies — spherical voids
- Shrinkage porosity: metal contracts while cooling without enough feed metal — irregular voids

Stress concentrators that reduce fatigue life. Detected by radiography (X-ray, CT scan).

Inclusions

Non-metallic particles (oxides, sulfides, silicates) trapped inside the metal during melting.

- Disrupt metal continuity throughout
- Common initiation sites for fatigue cracks in service

Managed by vacuum melting, good melt practice, and clean steel processing standards.

Cracks

Planar separations within the metal — the most dangerous defect class.

- Hot cracks: form at grain boundaries during solidification
- Cold cracks: form during cooling, heat treatment, or in service (fatigue, stress corrosion)

Stress concentrators that grow rapidly under cyclic load. Detected by dye penetrant, magnetic particle, UT, or radiography.

Delaminations & Segregation

Two structural defect types:

- Delaminations: separation between rolled/forged layers from elongated inclusions — detected by UT
- Segregation: uneven distribution of alloying elements from solidification — causes localized hardness and corrosion resistance variations

Both require careful process control and systematic inspection during manufacturing.

SLIDE 13 *Crystal Defects — Volume Defects*

NOTES

Grain Boundaries — Strengthening and Growth

Grain boundaries are high-energy regions that actively control strength, toughness, corrosion, and diffusion — not just where grains meet.

The Hall-Petch Relationship

$$\sigma_Y = \sigma_0 + k / \sqrt{d}$$

Yield strength increases as grain size (d) decreases — smaller grains mean more boundaries, and dislocations pile up sooner.

- When a dislocation reaches a grain boundary, it cannot cross — orientation mismatch blocks it
- Dislocations pile up, creating a back-stress that resists further deformation
- Smaller grain = shorter distance before hitting a boundary = more pile-ups = stronger
- One of the most powerful and practical strengthening tools in metallurgy

Grain Growth in Annealing

1. Recovery
Internal stresses relieved, no major microstructure change. Metal softens slightly but retains deformed grain shape.

2. Recrystallization
New strain-free grains nucleate and replace deformed grains. Strength drops, ductility fully recovers.

3. Grain Growth
New grains coarsen as boundaries migrate to reduce total energy. Control time and temperature to reach target grain size.

SLIDE 14 *Grain Boundaries — Strengthening and Growth*

NOTES

Diffusion — Atoms in Motion

Even in solid metal, atoms are not perfectly still — they vibrate constantly, and at elevated temperatures they have enough energy to jump between sites.

Substitutional Diffusion

Larger atoms move by jumping into neighboring vacancies — hopping from one lattice site to an adjacent empty one.

Requires: a nearby vacancy AND enough thermal energy to squeeze through the gap between atoms.

Rate: Slow — vacancies are relatively rare and large atoms must squeeze between others.

Example: Alloying additions spreading through steel: Cr, Ni, Mo homogenization during annealing

Interstitial Diffusion

Small atoms move through the gaps between host atoms without displacing them — flowing through the interstitial network.

No vacancy required — gaps are everywhere, so this is much faster than substitutional diffusion.

Rate: Fast — gaps are abundant, and small atoms can move easily even at lower temperatures.

Example: Carbon in iron (carburizing, case hardening), nitrogen in steel (nitriding)

Temperature Effect: Diffusion rate increases exponentially with temperature — a small temperature rise causes a dramatic increase in atom movement. This is why heat treatment temperatures must be controlled precisely.

SLIDE 15 *Diffusion — Atoms in Motion*

NOTES

Core Questions — Answers

1 *What is the difference between strength and toughness?*
Strength is the peak stress before yielding or fracture. Toughness is total energy absorbed — the area under the stress-strain curve. A material can be strong but brittle (high-carbon steel, glass): high peak stress, negligible plastic zone, low toughness.

2 *What happens at the yield point at the atomic level?*
Dislocations overcome their pinning obstacles and begin moving irreversibly through the crystal. Below yield, bonds stretch elastically and recover. At yield, atomic planes shift permanently — plastic deformation begins and cannot be undone by removing the load.

3 *Why does cold work increase strength AND reduce ductility simultaneously?*
Cold work multiplies dislocation density enormously. Dislocations tangle and block each other — work hardening — requiring more stress to continue deformation (higher strength). But with so many dislocations already present, little capacity for further plastic flow remains (lower ductility).

4 *Fine-grained vs. coarse-grained steel: which is stronger and why?*
Fine-grained. Hall-Petch: $\sigma_y = \sigma_0 + k/\sqrt{d}$. More grain boundaries per unit volume block dislocation movement at each boundary — each crossing requires additional stress. Finer grains = more barriers = higher yield strength.

5 *Shaft fails below yield strength under cyclic loading. What is it?*
Fatigue failure. Cyclic loading initiates tiny cracks at stress concentrations (surface defects, notches) that grow with each cycle until catastrophic fracture — often well below the static yield strength. The relevant property is the endurance limit.

SLIDE 16 *Core Questions — Answers*

NOTES

Key Takeaways

1 Stress and strain are the engineer's language

Stress ($\sigma = F/A$) is internal force per area. Strain ($\varepsilon = \Delta L/L_0$) is deformation relative to original size. The yield point — the elastic-to-plastic transition — is the most important value in structural material selection.

2 Defects are features, not failures

Point, line, planar, and volume defects each affect metal differently. Dislocations enable ductility — eliminate them and the metal becomes brittle. Real engineering metals are defined by their defect structure.

3 Grain size is the most practical lever

Hall-Petch: $\sigma_y = \sigma_0 + k/\sqrt{d}$ — making grains smaller makes metal stronger. Grain size is controlled through alloy additions and thermal processing (annealing temperature and time).

4 Diffusion drives heat treatment

Atoms move through solids at rates exponential with temperature. Carburizing, nitriding, and homogenization all rely on controlled diffusion — this is why heat treatment temperatures are precise.

5 Safety is not optional!

PPE is mandatory in the shop — no exceptions. We work with heat, sharp edges, and hazardous materials every lab session. Protect yourself and your classmates.

SLIDE 17 *Key Takeaways*

NOTES

Resources & What's Next

Week 2 Resources

- Moniz Ch. 3 — Mechanical Properties of Metals
- Kalpakjian Ch. 2 — Fundamentals of Mechanical Behavior of Materials

Visualizers

- Stress-Strain Curve (Interactive)
- Hot vs Cold Working & Work Hardening (Interactive)

Coming Up — Week 4

Carbon & Alloy Steels

- Week 3 is Labor Day — no class
- Iron vs. steel — what carbon does to the lattice
- AISI/SAE four-digit classification system
- Plain carbon vs. alloy steel grades and alloying elements

Unit 1 Quiz covers Weeks 1 – 2 · Review bonding, crystal structures, and the stress-strain curvees

SLIDE 18 *Resources & What's Next*

NOTES